Thomas Ajai

New York City | (646) 337-8401 | tl3444@columbia.edu | www.linkedin.com/in/thomas-ajai | www.github.com/thomasajai

EDUCATION

Columbia University **Aug 2025 - Dec 2026**
MS, Computer Science (GPA: 4.04) *New York City, NY*

College of Engineering Trivandrum **Aug 2019 - Jun 2023**
B. Tech, Computer Science and Engineering (GPA: 9.52/10) *Trivandrum, IN*

PROFESSIONAL EXPERIENCE

UST | *AI Data Engineer Intern* **Aliso Viejo, CA (Remote) | May 2026 – Aug 2026**

- Engineered multi-tenant Workspace isolation for an AWS data and AI platform, propagating tenant and lakehouse scope across catalog, modeling, pipelines, background workers, AI applications, and AWS resource authorization.
- Built and hardened data platforms across Snowflake, PostgreSQL, Redshift, Amazon S3 Tables, Glue/Athena, and OpenSearch; added reusable data-source connections, schema profiling, drift detection, lineage, and Bedrock semantic search.
- Developed durable AWS provisioning and agentic workflows with persistent execution state, resumable workers, heartbeat recovery, idempotent resource creation, and failure recovery for long-running cross-account infrastructure operations.
- Built a context-aware AI assistant and centralized Amazon Bedrock observability layer, persisting conversation context and attributing token usage and cost by tenant, user, model, feature, and Workspace.

Providence India | *Data Engineer* **Hyderabad, India | Jul 2023 – Jul 2025**

- Built end-to-end ETL pipelines using Alteryx and Azure Data Factory, integrating Epic, Cerner, and Athena EHR data across more than 53M healthcare records.
- Migrated production data workflows to Snowflake SQL and orchestrated stored procedures through Azure Data Factory, reducing ETL runtime by 80% while improving analytics reliability.
- Developed Power BI dashboards tracking 25+ clinical, telephony, and patient-experience KPIs, including call volume, abandonment, OSAT, and first-inquiry resolution across organizational hierarchies.
- Automated production workflow monitoring and data-quality validation, sending scheduled run-status notifications to product owners through Power Automate and webhooks for faster issue visibility.

SELECTED PROJECTS

Industrial Time-Series Foundation Model Agent Optimization | PyTorch, TinyTimeMixer, CUDA, torch.compile

- Optimized the TSFM MCP agent in IBM AssetOpsBench for predictive-maintenance workflows, benchmarking TinyTimeMixer variants with approximately 1M and 5M parameters.
- Profiled CPU, CUDA kernel, matrix-multiplication, memory-transfer, and compilation overhead to isolate inference bottlenecks across repeated industrial sensor forecasting and anomaly-detection workloads.
- Implemented BF16 model preloading, torch.compile with reduced-overhead execution, and inference caching to reduce repeated model setup and improve GPU utilization across agent calls.

3D Reconstruction with Gaussian Splatting | PyTorch, 2DGS, 3DGS, CUDA

- Trained and evaluated 2D and 3D Gaussian Splatting pipelines on DTU Scan 105, comparing reconstruction quality using PSNR, SSIM, and LPIPS.
- Achieved 29.26 PSNR, 0.901 SSIM, and 0.212 LPIPS with 2D Gaussian Splatting after 30,000 training iterations on the held-out evaluation views.
- Analyzed rendering quality, convergence behavior, and representation differences between 2DGS and 3DGS, using quantitative metrics and rendered outputs to diagnose reconstruction performance.

LLM Post-Training for Mathematical Reasoning | PyTorch, LLaMA 3.2, LoRA, REINFORCE

- Post-trained LLaMA 3.2 1B on GSM8K using LoRA supervised fine-tuning and REINFORCE, improving grade-school mathematics exact-match accuracy from approximately 3% to 24%.
- Implemented custom training loops, reward functions, prompt masking, and formatting-constrained generation using PyTorch and Hugging Face Transformers for reinforcement-learning-based optimization.
- Built an automated evaluation pipeline for numeric answer extraction, tolerance-aware exact-match scoring, and reproducible benchmarking across generated mathematical reasoning responses.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Go, C++, JavaScript, TypeScript
- **AI & ML:** PyTorch, Transformers, LoRA, LLM Post-Training, Computer Vision, CUDA, Amazon Bedrock, Hugging Face
- **Data & Backend:** PostgreSQL, Snowflake, PySpark, FastAPI, SQLAlchemy/SQLModel, Azure Data Factory, Alteryx
- **Cloud & DevOps:** AWS (Glue, Athena, S3 Tables, Lake Formation, IAM, ECS Fargate), Docker, CloudFormation, GitHub CI/CD, Linux